

M002 Extension 4 — Complete the Maze

Topic: Add Two AND Conditions · **Course:** Maze Madness · **Level:** Extension (Advanced) · **Time:** about 55 minutes

The agent knows **four rules** to follow a redstone trail through a 3D maze. The solver is **not finished**: at two junctions two signals are on at once, so one signal is not enough and the agent stops. Add **two AND rules** of your own — **you** decide which two signals each junction needs. Then drive the agent to the goal.

Chat: l turn left, r turn right, run start solver, rl bring agent back.

1 · Read the Four Rules

Each rule checks **one** signal and does **one** action.

```
if agent.detect(REDSTONE, LEFT):
    agent.turn(RIGHT_TURN)
elif agent.detect(REDSTONE, RIGHT):
    agent.turn(LEFT_TURN)
elif agent.detect(REDSTONE, DOWN):
    agent.move(UP, 1)
elif agent.detect(REDSTONE, UP):
    agent.move(DOWN, 1)
```

Fill the table from the code. You will reuse these actions.

Signal	Action
redstone left	
redstone right	
redstone below (down)	
redstone above (up)	

A left signal makes the agent turn *right*; a signal below makes it move *up*. Why might the maze pair a signal on one side with an action to the other?

2 · Find the Two Stuck Junctions 🔍

Most junctions have **one** signal on, so one rule fires. Two junctions have **two** signals on at once — there the four rules are not enough and the agent stops.

Type **run**. Watch where it stops. Walk over and look around the agent.

Tick the signals on at each stuck junction.

	left	right	below	above
Junction 1				
Junction 2				

Only the first matching **if / **elif** branch runs on each pass. Why does that leave the second signal unused?**

3 · AND means BOTH

An **AND** is true **only when both** halves are true.

```
if agent.detect(REDSTONE, LEFT) and agent.detect(REDSTONE, DOWN):  
    ...
```

(Example only — yours may differ.) Left on but NOT below: is the whole condition true? Below on but not left?

Look at Junction 1 in your table. Write the AND condition true only at that spot, then the action for each signal in the order you want them.

4 · Add the Two New Rules

Open the **M002 Complete the Maze** world. Find the four rules. Add **two** **elif** branches, one per stuck junction, using the signals **you** found.

Junction 1 — write the whole branch: the **AND** condition, then the action for each signal.

```
elif agent.detect(REDSTONE, _____) and agent.detect(REDSTONE, _____):  
    agent._____(_____)  
    agent._____(_____)
```


Junction 2 — write the whole branch from scratch.

Both new rules must be checked *before* the four single-signal rules. Why? What happens at a stuck junction if a plain single-signal rule is checked first?

5 · Spot the Bug 🐛

Each block is broken. Rewrite it, then explain the fix. (Signals here are examples.)

Bug A — should fire **only when both** signals are on, but fires when **either** one is, so it triggers at wrong spots.

```
elif agent.detect(REDSTONE, LEFT) or agent.detect(REDSTONE, DOWN):  
    agent.turn(RIGHT_TURN)  
    agent.move(UP, 1)
```

Write the fixed code:

Why was it wrong?

Bug B — a junction needs **two** actions, but this rule does one, so the agent solves half and sticks again.

```
elif agent.detect(REDSTONE, RIGHT) and agent.detect(REDSTONE, UP):  
    agent.turn(LEFT_TURN)
```

Write the fixed code:

Why was it wrong?

run

Each of the four rules checked ...

The agent got stuck because one signal was not ...

The two signals at the first stuck junction were ... and ...

My AND rule was true only when both were ...

I put my two new rules before the old rules because ...

The hardest part was ...

7 · Plan Your Walkthrough Video 🎥

Film on your phone (or a parent's) while the agent solves the whole maze. Teach it like the viewer has never seen it. Try these words: **detect, redstone, AND, both, condition, turn, move up, move down.**

1. Show the maze and point out the redstone signals.
2. Type **run** and show the agent following the trail.
3. Read one AND rule and say which two signals must be true.
4. Show the agent passing a stuck junction.
5. Show it reaching the goal.

Write what you will say. Plan it here first — read from it while filming.

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.